

Project Proposal: Design And Implementation Of a Salt Powder And Packaging Machine With The Auger Filler System

Section A: Executive Summary & Introduction

We propose to design and build an automated salt powder packaging line that uses OPP/CPP plastic film and an ESP32-based control system. The system will form bags from film, fill each with a precise salt dose, heat-seal the bag, and cut it, achieving at least 10 bags per minute throughput and the approximate weight will be between the following two Values 500 gram - 2000 gram. This project builds on modern packaging technology For Example, Vertical Form Filling Seal (VFFS) Machines Create Bags From Film Rolls and Adapts it to our Scale and Specifications. The Machine will include a film unwind, forming tube, filling auger, load cell weigh scale, heating sealing jaws, and an ESP32 microcontroller orchestrating sensors and motors. All key design choices (material type, components, control) reflect discussions to date and align with the stated objectives.

1.1.1 Key Objectives

- **Automated Bagging:** Build a VFFS machine for salt powder, using OPP/CPP film as the packaging material (these polypropylene films offer high strength and moisture resistance).
- **Throughput Target:** Achieve a production speed of ≥ 10 bags/min (600 bags/hr), matching ~Medime industrial baggerr.
- **Weight Accuracy:** Pack a precise target weight per bag (e.g. 1 kg of salt) with minimal variance, using a load-cell for feedback.
- **Control System:** mplement an ESP32 microcontroller to automate sequencing (film feed, weighing, sealing) and user interface (start/stop controls).
- **Cost Constraint:** Complete the design within a budget of \ \$1500 USD, covering all materials, electronics, and assembly.

1.1.2 Production Line Features

- **Film Handling:** A roll of OPP/CPP film is unwound and formed into a tube via a forming collar. OPP (uniaxially oriented PP) film is selected for its clarity and strength; CPP (cast PP) may be used as a sealable inner layer due to its excellent moisture barrier.

- **Filling System:** A calibrated auger dispenses salt into each forming bag. A load cell scale under the fill zone measures the dispensed weight, ensuring each pouch meets the target weight.
- **Sealing Mechanism:** Heated sealing bars (powered by a controlled heater) will vertically seal the film edges and horizontally seal/cut each bag. The high-melting-point properties of polypropylene allow reliable heat-sealing.
- **Actuators & Sensors:** Stepper motors (with drivers) advance the film and operate to rotate the blades and auger, while sensors (limit switches or photoeyes) detect film position. Relays will switch the heater and motor drivers under ESP32 control. This reflects the block diagram approach of similar systems.
- **Control Electronics:** The ESP32 board reads inputs (start button, sensors, load cell ADC) and executes the bagging sequence in software. It controls outputs (motor drivers, heater relay) to coordinate tasks automatically. An display(HMI) or connectivity module provide status.
- **Frame and Hygiene:** All product-contact parts will use safe materials (e.g. stainless steel hoppers,..), and the frame will support easy cleaning. The enclosed design minimizes contamination.

1.1.3 Financial Overview

- **Budget:** The total budget is set at \ \$1500 USD. This must cover all components: film stock, mechanical parts (rollers, frame), motors, sensors (load cell, photo-sensors), ESP32 board, power supply, wiring, and labor.
- **Cost Allocation:** Electronics (ESP32, drivers, sensors) cost a few hundred dollars; mechanical elements (motors, frame, sealing bars) use the remainder.

1.1.4 Summary

In summary, this proposal outlines an integrated packaging line for salt powder that meets the previously discussed requirements. By combining established VFFS technology with modern IoT control, the system will reliably produce at least 10 sealed bags per minute, each with a precise salt dosage. The design is consistent and compatible with our earlier design discussions (film type OPP/CPP, use of ESP32, component list, etc.) and is structured to deliver the stated outputs within the specified budget In Sha" a Allah.

Section B: Background and Problem Analysis

2.1 Historical Background

Automated packaging machinery has evolved significantly since the mid-20th century. The first vertical form-fill-seal (VFFS) machine was introduced in 1959, using a draw-down belt mechanism to form, fill, and seal bags, a principle still used today. Over the decades, improvements in film materials accompanied the

machinery advances. Polypropylene films (like OPP/CPP) became popular since the 1970s for food packaging due to their superior clarity and moisture barrier. Biaxially-oriented PP (BOPP) and uniaxially-oriented PP (OPP) films allowed transparent, high-strength bags. Modern machines now reach extremely high speeds (hundreds of bags per minute), though our design targets a modest 10 bags/min to suit project scope. This history of incremental innovation from basic film extruders to today's servo driven VFFS line provides the foundation for our design.

2.2 Problem Definition

The local salt-packaging sector in Yemen faces a persistent capability gap: small and medium producers largely depend on imported packaging machinery that is costly to acquire, difficult to maintain, and frequently incompatible with local operating conditions (dusty environments, variable grid power, and specific powder characteristics). As a result, packing operations are often manual or semi-automated, which produces low and inconsistent throughput, poor weight accuracy, frequent production stoppages for maintenance or spare parts, and higher unit packaging costs than achievable with localized automation. These constraints directly lower factory productivity, increase product loss (under-/over-filling and film waste), and reduce competitiveness of domestic salt producers in local and regional markets.

Key problems include:

- **Low Throughput:** Current methods often produce far less than the desired target (10 bags/minute), thus limiting production capacity and revenue.
- **Weight Inaccuracy:** Human filling introduces significant variance in bag weight. Lack of precise dosing can cause underfill or overfill.
- **Environmental and product-handling issues:** Powdered salt tends to bridge, rat-hole, and create airborne dust; without purpose-designed hoppers, anti-bridging measures, and dust extraction, feed reliability and worker safety suffer.
- **Energy vulnerability:** Many imported solutions assume stable grid power and higher energy budgets; local facilities often need lower-power designs and compatibility with small solar or UPS backups.

These issues highlight the need to add an external tank and an auger between the external tank and the internal hopper to move the salt powder, higher production throughput of salt powder, and precise filling and sealing. For example, similar automation efforts have produced multiplicative efficiency gains: an ESP32-controlled cup-closing machine increased output from about 50 cups per hour to 300 cups per hour, illustrating how electronics can significantly improve productivity.

2.3 Overall and Specific Objectives

- **Overall Objective:** Develop a fully automated form-fill-seal packaging machine for salt powder using OPP/CPP film, capable of high-speed, accurate bagging.

- **Specific Objectives:**
- Achieve a steady output of ≥ 10 bags/min with clean seals.
- Use an **ESP32 microcontroller** to automate the sequence and allow easy adjustments.
- Integrate a **precision weighing system** (load cell) to control fill quantity.
- Employ durable, food-safe materials (stainless steel, plastic) in all contact parts.
- Ensure the total build cost does not exceed \ \$1500.

2.4 Targeted Activities and Outputs

Planned activities:

- **Design & Engineering:** Create mechanical and electrical schematics. Select components (film feed, motors, sensors, etc.) based on our discussions.
- **Procurement:** Acquire all required parts (OPP/CPP film rolls, ESP32 board, motor drivers, sensors, heaters, framing materials).
- **Assembly:** Build the machine frame, install film unwind system, forming tube, conveyor, hopper, and sealing jaws. Wire the ESP32, connect drivers and relays.
- **Software Development:** Program the ESP32 to control the process (start/stop sequence, motor actuation, heater timing, weight feedback).
- **Testing & Calibration:** Run trials with salt to calibrate dosing and sealing temperature. Adjust timings and load-cell calibration for consistent bag weights.
- **Documentation:** Prepare user manual, maintenance guide, and the final report/presentation.

Expected outputs:

- A **working prototype** packaging line (in PDF proposal it will be documented and later in a physical build).
- Demonstrated **production capability** of ≥ 10 bags/min and correct weight per bag.
- Project report including design drawings, code listings, test data (e.g. consistency of bag weights).
- A final presentation or PDF summarizing the system (mechanical drawings, control diagrams, etc.).

2.5 Beneficiaries

- **Salt Producers/Manufacturers:** Gain a reliable, automated packing solution that boosts capacity and reduces labor costs.
- **End Consumers/Retailers:** Receive consistently filled, well-sealed salt bags, improving product quality and shelf life.
- **Engineering Team/Students:** Develop skills in mechatronics, IoT control, and packaging technology through this project.

- **Local Industry and Economy:** Encourages local innovation by providing a low-cost packaging machine, and can be adapted for other granular products.

2.6 Impact and Impact Pathways

The project's implementation will have several positive impacts:

- **Productivity:** Meeting 10 bags/min significantly increases output versus manual methods, effectively multiplying production rates (consistent with the ~6× improvement seen in similar automation[10]).
- **Quality Control:** Automated weighing ensures every bag meets target weight, reducing waste from under/over-filling.
- **Cost Efficiency:** Automated processes will lower long-term labor and material waste costs. Precise cutting and bagging minimizes film waste.
- **Knowledge and Technology Transfer:** Demonstrates how IoT controllers (ESP32) can be applied in manufacturing[2], which can inspire further automation projects.
- **Scalability:** Although designed for salt, the line's principles can extend to other powders/granules, multiplying its benefit in the food industry.

2.7 Relevance

This project is highly relevant to industry needs and educational goals. It aligns with modern **Industry 4.0** trends by integrating sensors and IoT control (ESP32) for real-time automation. Using OPP/CPP packaging is standard in the food sector[4], so our system meets actual market requirements. At the same time, the project fits within an **academic/engineering curriculum** by covering mechanical design, electronics, and control software. The \$1500 budget makes it a feasible small-scale venture, showing how even constrained resources can realize advanced manufacturing solutions.

Section C: Methodology and Timeline

3.1 Methodology

We will employ a classic **Vertical Form-Fill-Seal (VFFS)** approach[1][3]. First, the film is unwound from a roll and guided through rollers to a forming collar. A tube is created as the film wraps around a vertical form tube, and the overlapping edges are sealed vertically by a heated bar[3]. The ESP32-based controller then activates the lower sealing jaw to form the bottom of the bag.

Next, salt is dispensed: an actuator (e.g. a stepper-driven auger) feeds salt into the newly-formed pouch. A load cell under the fill chute measures the weight in real time, and the ESP32 uses this feedback to stop filling precisely at the target weight. Once filled, a heated horizontal sealer closes the top and simultaneously cuts the bag off[3]. Sensors (photodetector or limit switches) and timed control ensure proper film indexing. Throughout, the ESP32 reads inputs and controls outputs

(motors, relays) as outlined in a typical automation block diagram[2]. The sequence of operations is: *Start* → *Feed Film* → *Seal Bottom* → *Feed Salt* → *Seal & Cut* → *Dispense Bag*, looping continuously. Each mechanical and electrical component is selected and tested to meet the specifications (e.g. motor torque for film feed, heater wattage for timely seal).

Key methodological steps:

- **Mechanical Design:** Draw and fabricate the machine frame, film unwind shaft, forming tube, feed conveyor, and seal jaws. Ensure adjustability for bag size.
- **Weighing System:** Integrate a calibrated load-cell under the filling area. Use an amplifier and ADC to interface its signal to the ESP32.
- **Actuation:** Use stepper or DC motors with drivers to advance film and drive any conveyors. Use relays or MOSFETs to power the sealing heater(s).
- **Control Software:** Write the ESP32 firmware (in Arduino/C++) to manage the state machine, read sensors, and actuate devices. Include safety checks and an emergency stop.
- **Testing & Iteration:** Perform iterative tests: verify bag integrity (seal strength), weight accuracy (adjust auger time or motor steps), and consistent cycle timing. Optimize code parameters (e.g. heater dwell time, motor speed).

Preliminary Timeline

1. **Weeks 1–2 (Design Phase):** Finalize design drawings for mechanical parts and electrical schematics. Define component list and order parts.
2. **Weeks 3–4 (Procurement):** Purchase all hardware (film, motors, sensors, ESP32 kit, power supply, wiring, frame materials).
3. **Weeks 5–7 (Assembly):** Build the frame and install mechanical elements (shaft, forming tube, conveyor). Mount motors, load cell, and sealing jaws.
4. **Weeks 8–9 (Electrical Integration):** Wire the ESP32, motor drivers, relays, and sensors. Assemble control panel (buttons, indicators).
5. **Weeks 10–11 (Software & Testing):** Program the ESP32 control logic. Conduct trial runs with salt, calibrate the load cell for correct weight, adjust motor timings, and verify sealing.
6. **Week 12 (Finalization):** Refine the system for reliability, perform final validation runs (e.g. sustained 10 bag/min), and complete documentation and presentation materials.

Section D: Conclusion and References

4.1 Conclusion

In conclusion, this proposal outlines a **comprehensive plan** for an automated salt powder packaging line that meets all previously discussed requirements. By leveraging proven VFFS principles[1][3] and modern ESP32-based automation[2][10], the design achieves high speed and precise dosing within the \$1500 budget. The machine's modular design (film feed, form-fill-seal, control)

allows straightforward implementation and future improvements. All key steps – from mechanical fabrication to electronic control and testing – have been mapped out in the methodology and timeline. Upon completion, the system will deliver uniform, sealed salt bags at the target rate, demonstrating a successful translation of our engineering discussions into a working solution.

4.2 References

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